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Palladium-Catalyzed Decarboxylation of Allenyl 3-Oxoalkanoates: An Efficient Synthesis of 3,4-Allenyl Ketones

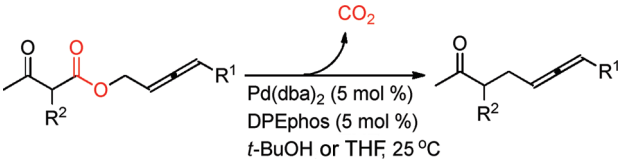
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ABSTRACT



An efficient synthesis of 3,4-allenyl ketones via the Pd-catalyzed decarboxylative coupling of the readily available 3-oxoalkanoates is reported. The C−C bond forming reaction occurs under mild conditions producing CO_2 as the only byproduct.

Allenyl ketones are particularly attractive due to the complementary reactivity of the carbonyl and allenyl groups and used as versatile intermediates in various reactions,[1] such as electrophilic additions,[2] nucleophilic additions,[3] transition metal-catalyzed cycloisomerizations,[4] and electrochemical reductive cyclization,[5] etc. Thus, the development of new methods for the synthesis of allenyl ketones is of high interest. On the other hand, decarboxylative coupling reactions[6] have become a powerful method for the construction of C−C

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(1) (a) *Modern Allene Chemistry*; Krause, N., Hashmi, A. S. K., Eds.; Wiley-VCH: Weinheim, 2004; Vols. 1, 2. (b) Ma, S. *Acc. Chem. Res.* **2003**, *36*, 701. (c) Trost, B. M.; Urabe, H. *J. Am. Chem. Soc.* **1990**, *112*, 4982. (d) Keck, G. E.; Kachensky, D. F. *J. Org. Chem.* **1986**, *51*, 2847.

(2) (a) Nagao, Y.; Lee, W. S.; Jeong, I. Y.; Shiro, M. *Tetrahedron Lett.* **1995**, *36*, 2799. (b) Hashmi, A. S. K.; Schwarz, L.; Bolte, M. *Tetrahedron Lett.* **1998**, *39*, 8969. (c) Marshall, J. A.; Robinson, E. D. *J. Org. Chem.* **1990**, *55*, 3450. (d) Marshall, J. A.; Wang, X. *J. Org. Chem.* **1991**, *56*, 960.

(3) (a) Marshall, J. A.; Tang, Y. *J. Org. Chem.* **1993**, *58*, 3233. (b) Zhang, C.; Lu, X. *Tetrahedron Lett.* **1997**, *38*, 4831. (c) Ma, S.; Shi, Z.; Li, L. *J. Org. Chem.* **1998**, *63*, 4522. (d) Ma, S.; Li, L.; Xie, H. *J. Org. Chem.* **1999**, *64*, 5325.

(4) (a) Marshall, J. A.; Wang, X. *J. Org. Chem.* **1992**, *57*, 3387. (b) Marshall, J. A.; Bartley, G. S. *J. Org. Chem.* **1994**, *59*, 7169. (c) Hashmi, A. S. K.; Ruppert, T. L.; Knöfel, T.; Bats, J. W. *J. Org. Chem.* **1997**, *62*, 7295. (d) Hashimi, A. S. K.; Schwarz, L.; Choi, J.-H.; Frost, T. M. *Angew. Chem., Int. Ed.* **2000**, *39*, 2285. (e) Hashmi, A. S. K. *Angew. Chem., Int. Ed. Engl.* **1995**, *34*, 1581. (f) Ma, S.; Zhang, J. *Chem. Commun.* **2000**, 117. (g) Ma, S.; Li, L. *Org. Lett.* **2000**, *2*, 941. (h) Ma, S.; Yu, Z. *Angew. Chem., Int. Ed.* **2002**, *41*, 1775.

(5) Pattenden, G.; Robertson, G. M. *Tetrahedron Lett.* **1983**, *24*, 4617.

(6) (a) Rayabarapu, D. K.; Tunge, J. A. *J. Am. Chem. Soc.* **2005**, *127*, 13510. (b) Goossen, L. J.; Deng, G.; Levy, L. M. *Science* **2006**, *313*, 662. (c) Waetzig, S. R.; Tunge, J. A. *J. Am. Chem. Soc.* **2007**, *129*, 4138. (d) Weaver, J. D.; Recio, A., III; Grenning, A. J.; Tunge, J. A. *Chem. Rev.* **2011**, *111*, 1846.

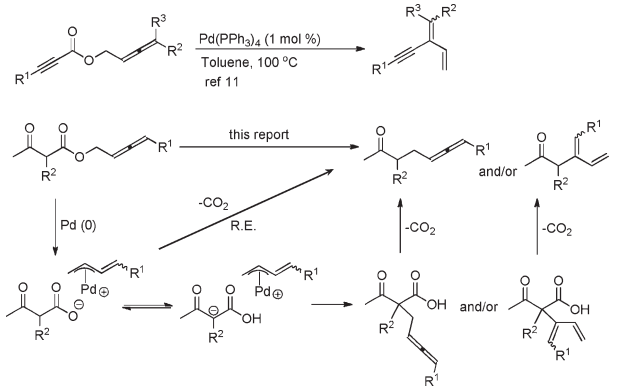
(7) (a) Tsuda, T.; Chujo, Y.; Nishi, S.-I.; Tawara, K.; Saegusa, T. *J. Am. Chem. Soc.* **1980**, *102*, 6381. (b) Shimizu, I.; Yamada, T.; Tsuji, J. *Tetrahedron Lett.* **1980**, *21*, 3199. (c) Tsuji, J.; Yamada, T.; Minami, I.; Yuhara, M.; Nisar, M.; Shimizu, I. *J. Org. Chem.* **1987**, *52*, 2988. (d) Burger, E. C.; Tunge, J. A. *Org. Lett.* **2004**, *6*, 4113. (e) Behenna, D. C.; Stoltz, B. M. *J. Am. Chem. Soc.* **2004**, *126*, 15044. (f) Tunge, J. A.; Burger, E. C. *Eur. J. Org. Chem.* **2005**, 1715. (g) Mohr, J. T.; Behenna, D. C.; Harned, A. M.; Stoltz, B. M. *Angew. Chem., Int. Ed.* **2005**, *44*, 6924. (h) Nakamura, M.; Hajra, A.; Endo, K.; Nakamura, E. *Angew. Chem., Int. Ed.* **2005**, *44*, 7248. (i) Waetzig, S. R.; Rayabarapu, D. K.; Weaver, J. D.; Tunge, J. A. *Angew. Chem., Int. Ed.* **2006**, *45*, 4977. (j) You, S.-L.; Dai, L.-X. *Angew. Chem., Int. Ed.* **2006**, *45*, 5246. (k) Burger, E. C.; Tunge, J. A. *J. Am. Chem. Soc.* **2006**, *128*, 10002. (l) Yeagley, A. A.; Chruma, J. J. *Org. Lett.* **2007**, *9*, 2879. (m) Sherden, N. H.; Behenna, D. C.; Virgil, S. C.; Stoltz, B. M. *Angew. Chem., Int. Ed.* **2009**, *48*, 6840. (n) Trost, B. M.; Xu, J.; Schmidt, T. *J. Am. Chem. Soc.* **2009**, *131*, 18343. (o) Chattopadhyay, K.; Jana, R.; Day, V. W.; Douglas, J. T.; Tunge, J. A. *Org. Lett.* **2010**, *12*, 3042.

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bonds, in part because these reactions may occur under neutral conditions and produce CO_2 as the only byproduct. Notable examples of Pd-catalyzed decarboxylative coupling reactions include decarboxylative alkylation,[7] decarboxylative Heck coupling,[8] aldol addition,[9] decarboxylative cross-coupling,[10] etc. In 2008, Chung et al. reported the synthesis of 2-alkynyl buta-1,3-dienes by decarboxylation of buta-2,3-dienyl 2′-alkynoates (Scheme 1).[11] We envisioned that such a strategy of decarboxylation of allenyl 3-oxoalkanoates would allow a convenient route to dienyl or allenyl ketones via an α-methylene π-allylpalladium intermediate (Scheme 1).[6d,7o] To the best of our knowledge, the synthesis of 3,4-allenyl ketones has not been well established: they may be prepared by the reactions of enolates[1c] or imines[5] with the alkylating agents, usually allenic chlorides or tosylates, using a strong base at low temperature. For the alcohol-oxidation approach, it is difficult to synthesize the corresponding 3,4-allenyl carbinols.[12] Thus, there is still a strong need for the development of a convergent synthesis of 3,4-allenyl ketones under mild conditions from readily available starting materials. Herein, we disclose such an efficient protocol to construct 3,4-allenyl ketones under mild conditions from the easily available allenyl 3-oxoalkanoates.

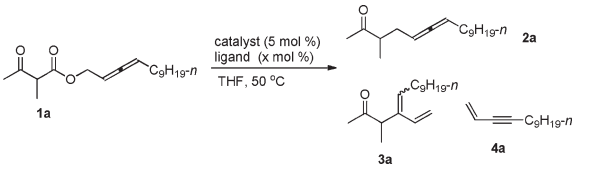
Scheme 1. Decarboxylation Reactions via an α-Methylene π-Allylpalladium Intermediate



To test our hypothesis, allenyl 3-oxoalkanoate **1a** was conveniently synthesized from readily available β-methylene-β-lactone[13] and the allenyl alcohol. When it was heated in THF at 50 °C in the presence of $Pd(PPh_3)_4$, the reaction failed to afford either the expected allenyl ketone **2a** or the 1,3-dienyl ketone **3a** (Table 1, entry 1). An initial catalyst screening revealed that $Pd(OAc)_2/PPh_3$, $Pd(dba)_2/PPh_3$, and $Pd(dba)_2$/LB-phos systems show similar results as $Pd(PPh_3)_4$ (Table 1, entries 2−4). However, the reaction afforded the allenyl ketone **2a** in 33% isolated yield as the only product by using $Pd(dba)_2$/dppf as the catalyst (Table 1, entry 5). Encouraged by this result, a series of ligands, such as dppe, binap, MeOBIPHEP, Xantphos, and DPEphos were screened for this transformation (Table 1, entries 6−10). Among them, DPEphos was shown to be the best with **2a** being formed in 59% isolated yield, and the formation of dienyl ketone **3a** was not observed. In this case, 1-vinylalkyne **4a** was also formed in 17% NMR yield (Table 1, entry 10).

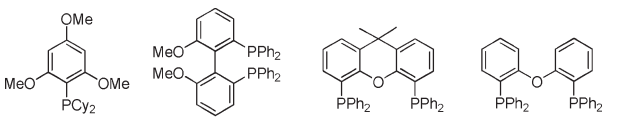
Using 5 mol % $Pd(dba)_2$ and 5 mol % DPEphos as the catalyst, subsequent comprehensive study on the solvent effect indicated that the reaction may proceed smoothly in all the tested solvents (Table 2): the reaction in *t*-BuOH afforded the best result with **2a** being formed in 66% isolated yield (Table 2, entry 10). By conducting the reaction at 25 °C, the yield of **2a** was improved to 77% although the reaction time was longer. In addition, the formation of **4a** was reduced to 6% NMR yield (Table 2, entry 11). Thus, the following optimized reaction conditions, i.e., 5 mol % $Pd(dba)_2$, 5 mol % DPEphos, *t*-BuOH, and 25 °C (Table 2, entry 11), were established for further study.

Table 1. Effect of Catalysts on the Decarboxylation of **1a**[a]



				NMR yield (%)[b]	
entry	catalyst	ligand (x/mol %)	time (h)	**2a**	**4a**
1	$Pd(PPh_3)_4$	–	21	complicated	
2	$Pd(OAc)_2$	PPh_3 (15)	21	complicated	
3	$Pd(dba)_2$	PPh_3 (15)	21	complicated	
4	$Pd(dba)_2$	LB-phos (15)	33	complicated	
5	$Pd(dba)_2$	dppf (5)	5.5	(33)[c]	–
6	$Pd(dba)_2$	dppe (5)	2.2	<14	–
7	$Pd(dba)_2$	binap (5)	2.3	30	19
8	$Pd(dba)_2$	MeOBIPHEP (5)	1.5	<48	16
9	$Pd(dba)_2$	Xantphos (5)	1.2	57	16
10	$Pd(dba)_2$	DPEphos (5)	0.5	66 (59)[c]	17

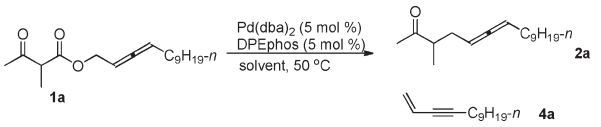
[a] Under argon, a mixture of **1a** (0.2 mmol) and the indicated catalyst in 2 mL of THF was stirred at 50 °C. [b] The yield was determined by ^1H NMR analysis using 1,3,5-trimethylbenzene as the internal standard. [c] The numbers shown in the parentheses are the isolated yields of **2a**.



(8) (a) Myers, A. G.; Tanaka, D.; Mannion, M. R. *J. Am. Chem. Soc.* **2002**, *124*, 11250. (b) Tanaka, D.; Myers, A. G. *Org. Lett.* **2004**, *6*, 433.
(9) Lou, S.; Westbrook, J. A.; Schaus, S. E. *J. Am. Chem. Soc.* **2004**, *126*, 11440.
(10) (a) Goossen, L. J.; Rodriguez, N.; Melzer, B.; Linder, C.; Deng, G.; Levy, L. M. *J. Am. Chem. Soc.* **2007**, *129*, 4824. (b) Becht, J.-M.; Catala, C.; Drian, C. L.; Wagner, A. *Org. Lett.* **2007**, *9*, 1781. (c) Goossen, L. J.; Melzer, B. *J. Org. Chem.* **2007**, *72*, 7473.
(11) Sim, S. H.; Park, H.-J.; Lee, S. L.; Chung, Y. K. *Org. Lett.* **2008**, *10*, 433.
(12) (a) Lotesta, S. D.; Hou, Y.; Williams, L. J. *Org. Lett.* **2007**, *9*, 869. (b) Zhang, X.; Ko, R. Y. Y.; Li, S.; Miao, R.; Chiu, P. *Synlett* **2006**, 1197.
(13) (a) Wilson, S. R.; Augelli, C. E. *Org. Synth.* **1990**, *68*, 210. (b) Wan, B.; Jia, G.; Ma, S. *Adv. Synth. Catal.* **2011**, *353*, 1763.

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Table 2. Effect of Solvent on the Decarboxylation of **1a**[a]

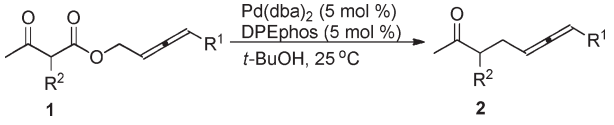


			NMR yield (%)[b]	
entry	solvent	time (h)	**2a**	**4a**
1	THF	0.5	66 (59)[c]	17
2	Dioxane	0.6	63	18
3	DME	0.6	62	18
4	Toluene	0.6	59	13
5	Benzene	0.6	58	15
6	CH_3CN	0.6	66	19
7	MTBE	0.6	59	15
8	DMF	0.6	65	18
9	EtOH	8	55	4
10	*t*-BuOH	0.6	70 (66)[c]	10
11[d]	*t*-BuOH	6.5	80 (77)[c]	6

[a] Under argon, the reaction was carried out with **1a** (0.2 mmol), $Pd(dba)_2$ (5 mol %), and DPEphos (5 mol %) in solvent (2 mL). [b] The yield was determined by ^1H NMR analysis using 1,3,5-trimethylbenzene as the internal standard. [c] The numbers shown in the parentheses are the isolated yields. [d] The reaction was carried out at 25 °C.

Then the substrate scope and generality of the reaction was investigated (Table 3): the reaction of allenyl 3-oxoalkanoates bearing a substituent at the terminal position of allenyl moieties **1a**–**1c** afforded the corresponding 3,4-allenyl ketones **2a**–**2c** as a mixture of ~1:1 diastereoisomers in 63–76% isolated yields; the reaction of **1d** with a terminal allene unit afforded the corresponding 3,4-allenyl ketone in 62% isolated yield; in addition, the reaction of **1a** proceeded smoothly to afford the allenyl ketone **2a** on a 1 g scale in 77% yield (Table 3, entry 2).

Table 3. Decarboxylative Coupling of 2-Oxoalkanoate **1**[a]



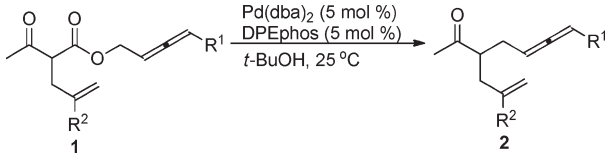
	1			
entry	R^1	R^2	time (h)	isolated yield of **2** (%)
1	n-C_9H_{19}	CH_3 (**1a**)	5	76 (**2a**)
2[b]	n-C_9H_{19}	CH_3 (**1a**)	3	77 (**2a**)
3	n-C_9H_{19}	Bn (**1b**)	4	65 (**2b**)
4	n-C_4H_9	Bn (**1c**)	4.5	63 (**2c**)
5	H	Bn (**1d**)	2.5	62 (**2d**)

[a] Under argon, the reaction was carried out with **1** (0.4 mmol), $Pd(dba)_2$ (5 mol %), and DPEphos (5 mol %) in solvent (2 mL). [b] The reaction was carried out on a 1 g scale.

(14) Ma, S. *Chem. Rev.* **2005**, *105*, 2829.

As is known, enallenes, allenynes, and bisallenes are useful substrates in organic synthesis.[14–17] The enallenes **2e**–**2i** could be obtained in moderate to good yields starting from 2-allylic substituted substrates **1e**–**1i** (Table 4). The diastereoisomer ratio of **2e**–**2g** is about 1:1.

Table 4. Synthesis of 2-Allylic Substituted 3,4-Allenyl Ketones by Decarboxylative Coupling of **1**[a]

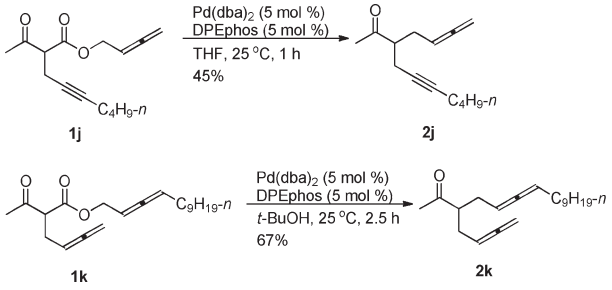


	1			
entry	R^1	R^2	time (h)	isolated yield of **2** (%)
1	n-C_9H_{19}	H (**1e**)	4	72 (**2e**)
2	cyclohexyl	H (**1f**)	5.5	74 (**2f**)
3[b]	*tert*-butyl	H (**1g**)	0.8	48 (**2g**)
4	H	Ph (**1h**)	7.5	70 (**2h**)
5	H	CO_2Me (**1i**)	3	60 (**2i**)

[a] Under argon, the reaction was carried out with **1** (0.4 mmol), $Pd(dba)_2$ (5 mol %), and DPEphos (5 mol %) in solvent (2 mL). [b] THF was used as the solvent.

Allenyne **2j** could be obtained in 45% isolated yield when THF was used as the solvent instead of *t*-BuOH (Scheme 2). In addition, the 2-(2,3-allenyl) substituted oxoalkanoate **1k** could be applied in the transformation affording the bisallene **2k** as a mixture of ~1:1 diastereoisomers in 67% yield (Scheme 2).

Scheme 2. Synthesis of Allenyne **2j** and Bisallene **2k** by Decarboxylative Coupling of **1j** and **1k**



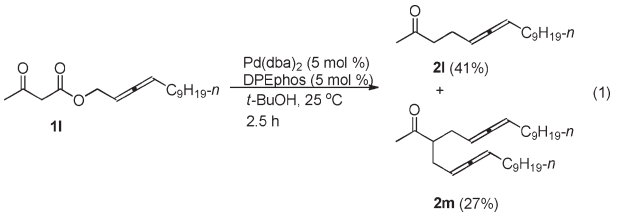
(15) For selected examples of cyclization of enallenes, see: (a) Teller, H.; Flügge, S.; Goddard, R.; Fürstner, A. *Angew. Chem., Int. Ed.* **2010**, *49*, 1949. (b) Alcarazo, M.; Stork, T.; Anoop, A.; Thiel, W.; Fürstner, A. *Angew. Chem., Int. Ed.* **2010**, *49*, 2542. (c) Zhao, J.-F.; Loh, T.-K. *Angew. Chem., Int. Ed.* **2009**, *48*, 7232.

(16) For selected examples of cyclization of allenynes, see: (a) Inagaki, F.; Sugikubo, K.; Miuashita, Y.; Mukai, C. *Angew. Chem., Int. Ed.* **2010**, *49*, 2206. (b) Siebert, M. R.; Osbourn, J. M.; Brummond, K. Y.; Tatillo, D. J. *J. Am. Chem. Soc.* **2010**, *132*, 11952. (c) Saito, N.; Tanala, Y.; Sato, Y. *Org. Lett.* **2009**, *11*, 4124.

(17) For selected examples of cyclization of bisallenes, see: (a) Ma, S.; Lu, P.; Lu., L.; Hou, H.; Wei, J.; He, Q.; Gu, Z.; Jiang, X.; Jin, X. *Angew. Chem., Int. Ed.* **2005**, *44*, 5275. (b) Jiang, X.; Cheng, X.; Ma, S. *Angew. Chem., Int. Ed.* **2005**, *45*, 8009. (c) Lu., L.; Ma, S. *Org. Lett.* **2007**, *9*, 2095.

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In addition, the reaction of 2-unsubstituented 3-oxoalkanoates **1l** afforded the monoallenylation product **2l** together with the diallenylation product **2m** in 41% and 27% isolated yields, respectively (eq 1).[18]



(1)

Unfortunately, reactions of the substrates **1m**−**1o** were complicated and failed to afford the corresponding decarboxylation products (Figure 1).

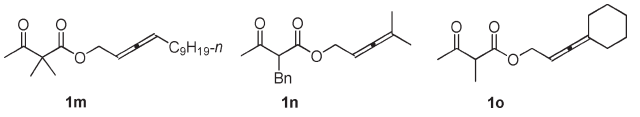


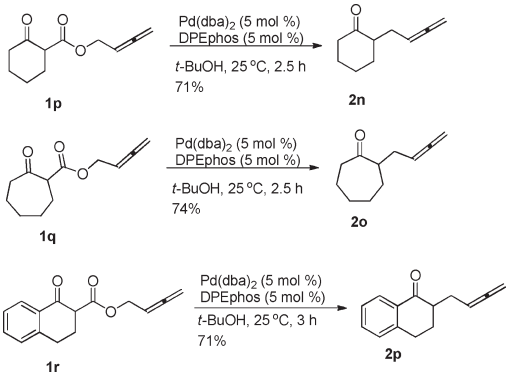
Figure 1. Other 3-oxoalkanoates tested for decarboxylation.

While the previous examples have focused on alkylation of acyclic 3-oxoalkanoates, the decarboxylative coupling of cylic substrates is also investigated. To our delight, the reaction of 3-oxoalkanoates **1p**−**1r** also afforded the corresponding allenyl ketones **2n**−**2p** in 71%, 74%, and 71% isolated yields, respectively (Scheme 3).

In conclusion, we have developed an efficient Pd-catalyzed decarboxylative coupling protocol for the synthesis of 3,4-allenyl ketones. The reaction occurs under mild conditions without external bases and produces CO_2 as the only byproduct. Due to the existence of both the carbonyl and allenyl moieties as well as the unsaturated C−C bond introduced by the substitution at the 2-position, this method will be useful for further application in organic synthesis. Further studies in this area including asymmetric synthesis of 3,4-allenyl ketones via this protocol are ongoing in our laboratory.

(18) The formation of diallylated product in decarboxylation allylation has been reported by Tsuji and co-workers; see ref 7b.

Scheme 3. Decarboxylative Coupling of Cyclic Substrates **1p**−**1r**



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Supporting Information Available. Analytical data for all products not listed in the text. This material is available free of charge via the Internet at http://pubs.acs.org.

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